

Supplement: Many-body gap protection of motional dephasing of an optical clock transition

Zhijing Niu, Vera M. Schäfer, Haoqing Zhang, Cameron Wagner, Nathan R. Taylor, Dylan J. Young, Eric Yilun Song, Anjun Chu, Ana Maria Rey, James K. Thompson¹

¹*JILA, NIST, and Department of Physics, University of Colorado, Boulder, CO, USA*

(Dated: February 10, 2025)

Experimental Setup

Our experimental system facilitates the continuous loading of strontium (Sr) atoms into a high-finesse optical ring cavity at a rate of 2.1×10^7 atoms per second [1]. The process begins with a heated Strontium-88 atomic beam source that generates a horizontal thermal beam. Atoms are initially cooled and collimated using a Zeeman slower and a two-dimensional magneto-optical trap (MOT) operating on the 461 nm $^1P_1 \rightarrow ^1S_0$ transition, enhancing the flux of atoms entering subsequent stages. Additional cooling is achieved using a 2D red MOT, 2D red molasses, and 3D red molasses based on the 7.5 kHz narrow-linewidth transition. An 813 nm optical lattice overlaps with the 3D red molasses to load atoms into the cavity mode. After loading, the atoms are released from the lattice, allowing them to move freely along the cavity axis. By varying the optical power in the 2D blue MOT stage, the number of atoms loaded into the cavity can be controlled without affecting the temperature of the atomic cloud, ensuring that the Doppler dephasing rate σ_D does not change as we vary the atom number.

Time-of-flight measurements

To measure the *rms* axial velocity of the atoms, which determines the predicted Doppler dephasing time scale, we perform time-of-flight measurements. The measurements begin by shutting off all laser cooling beams and allowing any untrapped atoms to fall away under gravity for 40 ms. We then turn off the 813 nm lattice beams in less than 200 ns. The cavity power at this wavelength then exponentially decays with a characteristic decay time of $\tau_L = 240$ ns. The lattice depth reaches less than 1 μ K in 1.2 μ s, negligible to the subsequent time of flight periods spanning out to 20 ms.

The cloud is allowed to ballistically expand along the cavity axis for a variable time t_f spanning 0 to 20 ms. We then use resonant 461 nm light to capture a fluorescence image of the atomic cloud which we fit to determine the rms cloud size $\sigma_z(t_f)$ size along the cavity-axis, as well as the center of the atomic cloud versus time. The falling of the center of the cloud with gravity is used to calibrate the image dimensions. The measured $\sigma_z(t_f)$ versus t_f is well described by the fit function $\sigma_z(t_f) = \sqrt{\sigma_0^2 + (\sigma_v t_f)^2}$, with the fitted rms axial velocity σ_v reported in the main text, and σ_0 capturing the initial rms cloud size along the cavity axis.

Calibration of χN

To characterize the exchange interaction strength $N\chi_-$, we exploit the fact that the full Hamiltonian of the system includes a previously neglected term $\hat{H}_c = \hbar \left[\Delta_c + \chi_- \left(\hat{N}_\downarrow - \hat{N}_- \right) + \chi_0 \left(\hat{N}_\downarrow - \hat{N}_0 \right) + \chi_+ \left(\hat{N}_\downarrow - \hat{N}_+ \right) \right] \hat{a}^\dagger \hat{a}$ where the operators $\hat{N}_\alpha$ are projection operators that count the number of atoms in a given atomic state. One sees that this Hamiltonian couples the inversion on each transition to the resonance frequency of the cavity. When all of the atoms are in the ground state, this results in a shift of the cavity frequency $\delta_c = \omega'_c - \omega_c = N(\chi_- + \chi_0 + \chi_+)$ from which we can solve for $N\chi_-$ since we know the coupling strengths and detunings. The cavity frequency shift δ_c is measured by applying a short pulse of drive light at the bare cavity resonance frequency and then detecting the subsequent ring down of the cavity field with the heterodyne detection. This is repeated with and without atoms to determine δ_c , and varying the drive light amplitude to ensure that the atoms are not significantly being excited out of the ground state. This calibration is performed at $\Delta_c = -2\pi \times 5$ MHz and at the highest atom number to in turn calibrate a fluorescence imaging system to scale to lower atom number.

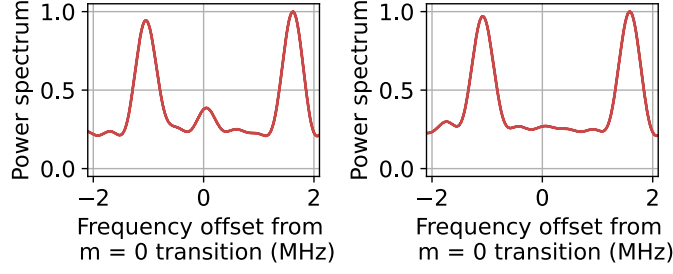


FIG. S1. The power spectra of the detected light from $t = 0$ to $2 \mu\text{s}$ (left) and $t = 2$ to $4 \mu\text{s}$ (right) show three and two discrete Fourier frequencies, respectively. The left plot shows three distinct emission peaks corresponding to the $m_j = -1$, $m_j = 0$, and $m_j = +1$ transitions. In contrast, the right plot shows only two distinct emission peaks corresponding to the $m_j = -1$ and $m_j = +1$ transitions.

Excitation of multiple excited states

As described in the main text, orienting the magnetic field as shown in Fig. 1d ensures nonzero coupling of all three transitions between ground and excited m_j levels to the relevant cavity mode. The $m_j = \pm 1$ states are simultaneously excited with Zeeman splitting δ_B by setting the drive frequency to $\omega_d = \omega_0$ and increasing its Rabi frequency.

Figure S1 presents the power spectrum of the measured cavity output after the excitation pulse. The spectrum computed for $t = 0$ to $2 \mu\text{s}$ (left) confirms excitation of the $m_j = 0$ transition. However, due to weaker cavity coupling and depletion of the ground-state population by other transitions, its coherence is not gap-protected. Consequently, the $m_j = 0$ peak is absent in the power spectrum for $t = 2$ to $4 \mu\text{s}$ (right).

The rapid decay of the $m_j = 0$ coherence stems from the absence of gap protection, unlike $m_j = \pm 1$, which remain protected. Two factors contribute to this: first, the ratio $\chi_0/\chi_- \approx 0.26$; second, significant ground-state depletion during excitation. The gap protection scale is determined by the total population in the combined ground (N_g) and excited (N_0) states. Using numerical simulations and short-time power spectrum peak heights, we estimate the energy gaps as 100 kHz for $m_j = 0$, 320 kHz for $m_j = -1$, and 260 kHz for $m_j = +1$.

Simulations indicate a dynamical phase transition between gap-protected and unprotected regimes at 210(40) kHz for $m_j = \pm 1$, where both transitions experience Doppler dephasing $\sigma_D = 87(9)$ kHz and magnetic field inhomogeneity $\sigma_B = 160(30)$ kHz, suggesting plausible gap protection. In contrast, the transition point for $m_j = 0$ occurs at 100(20) kHz, governed only by Doppler dephasing, as the $m_j = 0$ transition is insensitive to magnetic fields. This lack of gap protection leads to its rapid decay, explaining its absence in the long-time power spectrum in Fig. 4c.

Scaling of gap protection

We consider the scaling of gap protection against single particle dephasing. First, we will consider a cavity with infinite finesse $F \rightarrow \infty$ such that $\kappa \rightarrow 0$ and collective decay $\Gamma_s \rightarrow 0$. We also assume a simple two-level model. To be in a regime in which the cavity can be adiabatically eliminated, we require that the cavity detuning is larger than the collective vacuum Rabi splitting, so we take $\Delta_m = \alpha_V \sqrt{N} 2g$, where $\alpha_V \gg 1$ is a number of order 4 to 10. This minimum allowed detuning maximizes the interaction energy scale χN . For gap protection against motional dephasing, we want $\chi N = \alpha_G \sigma_D$ with $\alpha_G \sim 2$, i.e. the interaction scale at least two times larger than the rms dephasing rate σ_D . The maximum dephasing that can be suppressed is $\sigma_{max} = \frac{\sqrt{N}g}{2\alpha_V\alpha_G}$. For reference, neglecting collective and single particle decay, the infinite time coherence is numerically simulated and found to be $\frac{2}{N}|S_-| = 0.5, 0.79$ and 0.90 at $\alpha_G = 1.25, 2$, and 3 respectively.

We now ask if we turn down the finesse F to a finite value, when does the collective superradiant decay rate per particle for a Bloch vector at the equator equal the rate of dephasing $\Gamma_s = \sigma_{max}$, giving a minimum required finesse $F_{min} = \frac{\alpha_G}{8\alpha_V} \frac{\omega_{FSR}}{\sqrt{N}g}$. The free spectral range of the cavity is $\omega_{FSR} = 2\pi c/L$, where for a ring cavity L is the round trip length. This is the minimum cavity finesse needed to start to improve the coherence time relative to the original dephasing at rate σ_{max} . One can also ask what is the maximum useful finesse by considering when the collective decay per particle is equal to the single-particle spontaneous emission $\Gamma_s = \gamma$. One finds that spontaneous emission sets a maximum $F_{max} = \omega_{FSR}/4\gamma\alpha_V^2$, above which there is limited value in further increasing the finesse though there is no fundamental harm in doing so.

If the actual dephasing rate in the system is given by $\sigma_D \leq \sigma_{max}$, then one should operate at a larger cavity detuning $\Delta_D = \Delta_m (\sigma_{max}/\sigma_D)$ and one finds minimum and maximum finesse $F_{min,D} = F_{min} (\sigma_D/\sigma_{max})$ and $F_{max,D} = F_{max} (\sigma_D/\sigma_{max})^2$. The ratio of the actual finesse to the minimum finesse $F/F_{min,D}$ gives the ratio of the superradiance-limited coherence time relative to the original decoherence time scale $1/\sigma_D$.

We wish to concretely connect the above abstract Hamiltonian parameters to actual experimental parameters. For a two level system on axis of a ring cavity, the coupling is given by $g^2 = \frac{3}{8\pi} \frac{\lambda^2 c}{V_c} \gamma$, where the transition wavelength is λ , the cavity mode volume is $V_c = \pi w_0^2 L/2$, and w_0 is the smallest cavity mode waist (i.e. $1/e^2$ in intensity.)

In the experimental system here, the calculated maximum dephasing rate that can be enhanced is $\sigma_{max}/2\pi = 180$ kHz or a temperature of $170 \mu\text{K}$. The minimum and maximum finesse are $F_{min} = 43$ and $F_{max} = 4.2 \times 10^3$, to be compared to the actual finesse $F = 6.2 \times 10^3$. The above assumes $N = 10^6$, $\alpha_G = 2$ and $\alpha_V = 4$. Correcting for the geometric modification ($g_- = g \cos \theta_p / \sqrt{2}$) decreases $\sigma_{max}/2\pi$ to 120 kHz and increases F_{min} to 66 with a cavity detuning $\Delta_m/2\pi = 15$ MHz.

As an example for an optical transition narrower than the one utilized here, we consider the following: an atomic transition with $\gamma/2\pi = 10$ Hz, $\lambda = 700$ nm, mass of 88 u, a ring cavity with $L = 10$ cm, $w_0 = 100 \mu\text{m}$, $N = 10 \times 10^6$, $\alpha_G = 2$ and $\alpha_V = 4$. One finds $\sigma_{max}/2\pi = 21$ kHz which if the dephasing is due to Doppler broadening is a temperature of $2.3 \mu\text{K}$. The minimum finesse is $F_{min,D} = 3.9 \times 10^2 (\sigma_D/\sigma_{max})$ and the maximum finesse is $F_{max,D} = 3.1 \times 10^6 (\sigma_D/\sigma_{max})^2$.

[1] J. R. K. Cline, V. M. Schäfer, Z. Niu, D. J. Young, T. H. Yoon, and J. K. Thompson, Phys. Rev. Lett. **134**, 013403 (2025).